BlastSmart App:
Research on Surface Blasting Design and Fragmentation

2957271 | 2962051 | 2986976 | 2977301

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1 Abstract

The BlastSmart mobile application is being developed by Dwala Global Resources to digitise blast design calculations and improve the efficiency of blasting engineers in surface mining operations. This research provides preliminary research on two important functionalities of the BlastSmart App required during the first phase of the application's development: the Surface Blasting Design Calculator and the Fragmentation Calculator. The research reviews already existing blast design principles and fragmentation prediction methods to identify the parameters and calculations required for each module. The Surface Blasting Design Calculator is designed to determine burden, stemming length, and actual powder factor, while the Fragmentation Calculator predicts the mean fragment size. The findings provide the theoretical foundation for developing algorithms, flowcharts, and pseudocode that will support the implementation of these functionalities within the BlastSmart application

2 Background

Dwala Global Resources has initiated the development of the BlastSmart mobile application to modernise blasting operations through digital technology. The application is intended to replace traditional paper-based blasting manuals and manual engineering calculations with a digital platform capable of performing blast design calculations in real time. The project will be implemented in three phases, with the first phase focusing on the development of the Surface Blasting Design Calculator and the Fragmentation Calculator. The Surface Blasting Design Calculator will enable blasting engineers to calculate critical design parameters, including burden, stemming length, and actual powder factor, from user-supplied blast design inputs. The Fragmentation Calculator will estimate the mean fragment size after blasting, assisting engineers in evaluating blast performance. By automating these calculations, BlastSmart aims to improve calculation accuracy, reduce human error, enhance decision-making, and increase operational efficiency in surface mining.

3 Literature Review

Blast design is a critical process that determines the efficiency, safety, and cost-effectiveness of surface mining operations. Research shows that blast performance is primarily influenced by controllable parameters such as burden, spacing, stemming length, hole diameter, explosive density, and powder factor, together with geological characteristics of the rock mass (Khandelwal & Monjezi, 2013). The burden and spacing determine the volume of rock broken by each blast hole, while appropriate stemming improves the confinement of explosive gases, increasing energy utilisation and reducing flyrock, airblast, and toxic fumes (Oates & Spiteri, 2021). Furthermore, fragmentation prediction has become an essential component of blast design because it directly affects loading, hauling, crushing, and milling efficiency. The Kuz–Ram model remains one of the most widely adopted empirical models for estimating mean fragment size due to its simplicity and acceptable prediction accuracy in surface blasting applications (Khandelwal & Monjezi, 2013).

4 The design

The proposed solution was designed using a structured algorithmic approach to ensure that the required calculations could be implemented accurately and systematically. The problem was divided into two independent parts: the Surface Blasting Design Calculator and the Fragmentation Calculator. For each module, a step-by-step algorithm was developed in plain English to describe the logical sequence of operations before implementation. The algorithms begin by accepting user inputs, validating the input values, performing the required calculations, and displaying the results. The calculations are based on the equations implemented in the program, including:

Sub-drill depth $U = \frac{D}{4}$,
Explosive mass $M = \frac{\rho D^2}{1273}$,
Burden B = √((M × (BH − SL)) / (a × Tech_PF × BH)),
Stemming length $SL = 20D$

The fragmentation module estimates the mean size of the fragment using the Kuz–Ram equation:
$X_{50} = A(PF^{-0.8})M^{0.167}\left(\frac{115}{E}\right)^{0.633}$

where the rock factor (A), the technical powder factor (PF), the explosive mass (M) and the relative effective energy (E) are determined from user input. The algorithms were then converted into flowcharts using standard symbols to visually represent the program logic, calculations, and decision-making processes. This structured design ensured that the solution was logically organised and ready for implementation in Python.

5 Results

The developed pseudocode successfully models the logical sequence required to perform both the Surface Blasting Design Calculator and the Fragmentation Calculator. Execution begins by declaring all required variables before prompting the user to enter the necessary blasting parameters. Once the inputs are received, the pseudocode performs mathematical calculations sequentially to determine the sub-drill, explosive mass, stemming length, burden, and actual powder factor. Selection structures (IF, ELSE IF, and ELSE statements) are used to evaluate the user's rock type. Depending on whether the user selects Hard, Medium, Soft, or Very Soft, the corresponding technical powder factor is assigned. If an invalid rock type is entered, the pseudocode follows the ELSE branch and displays an ”Invalid Rock Type” message, preventing incorrect calculations. Similarly, in the Fragmentation Calculator, IF–ELSE IF statements are used to assign the appropriate rock factor and relative effective energy based on the selected rock type and explosive type (ANFO or TNT). Once valid values have been assigned, the pseudocode applies the Kuz–Ram fragmentation equation to calculate the mean fragmentation size (X) and displays the result. Throughout the solution, variable declarations, input statements, assignment statements, mathematical expressions, selection (decision-making) structures, and output statements are used to ensure that the calculations are performed logically and that only valid user selections are processed. The pseudocode therefore provides a clear and structured representation of the program logic, making it easier to implement the solution in a programming language while maintaining accuracy and readability.

6 Discussion of the Results

The developed algorithms, flowcharts, and pseudocode successfully address the requirements outlined in the project specification for the first phase of the BlastSmart application. The logical sequence ensures that all required user inputs are collected before calculations are performed, reducing the likelihood of invalid or incomplete results. The use of selection structures enables the program to assign the correct technical powder factor, rock factor, and explosive energy based on the user's selections, while mathematical expressions accurately compute the required blast design parameters and the predicted mean fragmentation size. The structured design also makes the solution easy to understand, test, and implement in Python. However, the current solution relies on predetermined empirical equations and predefined rock and explosive categories. It does not consider geological variability, moisture content, or discontinuities within the rock mass, which may influence actual blasting performance. These limitations highlight opportunities for future improvement.

7 Conclusion

This study successfully developed the preliminary design for two core functionalities of the BlastSmart application: the Surface Blasting Design Calculator and the Fragmentation Calculator. The research provided the theoretical basis for the calculations, while the algorithms, flowcharts, and pseudocode translated these requirements into a structured logical solution. The proposed design satisfies the functional requirements specified in the project brief by calculating burden, stemming length, actual powder factor, and mean fragmentation size. Overall, the project demonstrates how structured algorithm design can improve the accuracy, consistency, and efficiency of engineering calculations before software implementation.

8 Recommendations

Future development of the BlastSmart application should focus on expanding its functionality beyond the two calculators developed in this study. Additional modules, including the Underground Blast Design Calculator, Prediction Calculator, and Rapid Calculator specified in the project brief, should be incorporated. The application could also include a graphical user interface (GUI), data storage capabilities, and automatic report generation to improve usability. Furthermore, future versions could integrate machine learning models and real-time field data to improve fragmentation prediction and blast optimisation. Additional explosive types, rock classifications, and validation checks should also be incorporated to increase the flexibility and accuracy of the application for a wider range of mining conditions.

9 References

Khandelwal, M. & Monjezi, M., 2013. *Investigation of the blast fragmentation using the mean fragment size and fragmentation index.* **International Journal of Rock Mechanics and Mining Sciences**, 60, pp.1–9. Available at: https://doi.org/10.1016/j.ijrmms.2012.12.042

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